

Piecewise-Affine Local Volatility

Calibrating a Dupire surface from sparse quotes: model, PDE, gradients, and speed

Vol-Fitter Technical Notes, No. 04

Vol-Fitter

July 11, 2026

Abstract

This note documents the Vol-Fitter’s full local-volatility model: a continuous piecewise-affine (P_1) local-variance surface $\nu_\theta(\tau, x)$, calibrated directly to sparse vanilla and variance-swap quotes through the forward Dupire PDE as an implicit pricing map. Positivity of the nodal variances guarantees a positive surface on the grid’s hull; the implicit-Euler step matrix is an M -matrix, giving a discrete maximum principle and unconditional ℓ^∞ stability (Crank–Nicolson provably loses monotonicity on coarse grids); and butterfly/calendar cleanliness of the output is *measured* on the price grid, never assumed. Sensitivities are an analytic tangent Jacobian assembled from forward PDE sensitivities; the matrix-free Gauss–Newton solver avoids the dense SVD but consumes that same Jacobian (the true backward-PDE adjoint derived here remains future theory). Engineering — a compiled vectorized-Thomas march, a stall early-stop, warm seeds — cut the cold fit ~ 3 – $6\times$. Two case files: the six-day weekly that broke the strike grid ($108 \rightarrow 23.5$ vol bp) and the convex-wing hinge that flattened SPY until confined to the extrapolation tail ($25.7 \rightarrow 2.6$). Fresh evidence (commit [77d43fb](#), 2026-07-11): the synthetic quotes reprice to 0.6 vol bp with the local-vol surface itself recovered to 0.25 vol points on the quote-covered region; the product-path Bloomberg fit reads 2.4 (SPY) and 11.9 (NVDA) vol bp in-operator — and 15.9 / 56.3 on the *converged* reprice, the honest metric this revision adopts.

Contents

1	The production problem	3
2	The forward Dupire equation	3
3	The piecewise-affine surface	4
3.1	Parametrization	4
3.2	Grid design, and the short-dated stack	5
4	The discrete pricing map	6
4.1	Discretization	6
4.2	Why not Crank–Nicolson	8
4.3	Observation operator and no-arbitrage diagnostics	8
5	Variance swaps	8
6	What the data identifies, and the objective	9
6.1	What the data identifies	9
6.2	The objective	9
7	Sensitivities: tangent and adjoint	10
8	Solvers and the speed story	11
8.1	Where the time goes	11
8.2	What worked	11
9	Case files	12
10	Worked examples	13
10.1	Synthetic quote recovery — and what it does not show	14
10.2	The Bloomberg benchmark	14
11	What is original, and limitations	15
12	Traceability	15
A	Hyperparameter atlas	17
B	Performance notes	19
C	Reference implementation: Dupire extraction	19

1 The production problem

A slice model fits one expiry at a time and stitches expiries together with a calendar constraint. Production needs the opposite object too: one surface that prices *every* strike and expiry of a name simultaneously — for scenario transport (Note 12), variance swaps (Note 08), and as the projection target of the graph extrapolator (Note 14). The classical route — fit an implied surface, then *differentiate* it through Dupire’s formula — is numerically explosive exactly where production data lives: sparse wings, short tenors, stale quotes. Dividing two noisy numerical derivatives manufactures negative variances and spikes that no downstream PDE survives.

Invariant

1. Every optimizer iterate is a genuine, strictly positive local-variance surface *on the grid’s hull* — positivity is carried by the *parameters*, never policed by a penalty. (The controlled-linear put extrapolation *beyond* the hull is outside this guarantee; section 3.)
2. The pricing map never differentiates market data: quotes enter only as targets of a forward PDE solve.
3. The discrete prices obey a maximum principle — the scheme is monotone and unconditionally ℓ^∞ -stable. That is stability, not an arbitrage proof: butterfly and calendar cleanliness of the output are *measured* by explicit grid diagnostics, never assumed.
4. The short-dated rescues and speed levers are gated; each gate’s no-op behaviour is *individually* demonstrated (byte-identity where locked as such, stated tolerance where not) — there is no blanket byte-identity claim.

The pipeline in one line. Normalize and de-Americanize the quotes (Notes 05–06) → build the vertex grid and PDE lattice (section 3) → march prices and sensitivities forward (sections 4 and 7) → form the residual stack (section 6) → regularize and optimize under box bounds (section 8) → reprice on the *refined* operator (the converged metric) → report the convexity/calendar diagnostics.

The Vol-Fitter therefore follows the Andreasen–Huge philosophy [2]: *parametrize* the local variance directly, price *forward* through the Dupire PDE, and calibrate the parameters to the quotes.

Heuristic

Extracting local vol by $\sigma_{\text{loc}}^2 = \partial_T C / (\frac{1}{2} K^2 \partial_{KK} C)$ divides two noisy derivatives of a sparse surface — garbage in the wings. Calibrating *forward* never differentiates the data: it asks “what positive surface, run through the PDE, best reprices the quotes?” Positivity is imposed on the parameters, so every iterate is arbitrage-clean by construction — the LQD philosophy of Note 01, one dimension up.

2 The forward Dupire equation

Assumption 1 (Units and the two clocks). Work in forward-normalized units: $x = K/F_T$ is the normalized strike and $c(\cdot, x)$ the normalized call, so deterministic rates, borrow and dividends are absorbed into the forward (Note 06); the quotes are de-Americanized mids (Note 05). Three time objects are distinct: T is the *calendar* maturity; $\tau(T)$ is the monotone *event-weighted variance time* of Note 11 (the diffusion clock, equal to T when the event clock is off); and the model’s nodal

parameters θ are local variances *per unit variance time* — the PDE below marches in τ , and every maturity-like symbol in this note ($\Delta T = 0.01$, the SDE clock, the var-swap integral, the time vertices) lives on the τ axis. In calendar time the same dynamics read $\partial_T c = \frac{1}{2} \dot{\tau}(T) \nu(\tau(T), x) x^2 \partial_{xx} c$. The model is a pure diffusion: no jumps, so a genuine short-dated event spike is only approximated.

Writing local variance $\nu(\tau, x) = \sigma_{\text{loc}}^2(\tau, x)$ on the variance clock, the forward Dupire equation is

$$\boxed{\partial_\tau c(\tau, x) = \frac{1}{2} \nu(\tau, x) x^2 \partial_{xx} c(\tau, x), \quad c(0, x) = (1 - x)^+} \quad (1)$$

It is a forward parabolic PDE in (τ, x) with the payoff as initial data [1]. (Below we write a generic t or T for the marching variable where no confusion is possible; it always means τ .) Its structural virtue is that positivity of ν *propagates* both static-arbitrage conditions:

Proposition 1 (Positive local variance propagates no-arbitrage). *Let c solve (1) with $\nu \geq 0$ bounded and the payoff initial condition. Then for all $\tau > 0$: (i) $\partial_{xx} c(\tau, \cdot) \geq 0$ (no butterfly arbitrage); (ii) $\partial_\tau c(\tau, x) \geq 0$ — monotonicity of the normalized call at fixed $x = K/F_T$. Under deterministic carry this is equivalent to the familiar calendar-spread condition at fixed dollar strike K : the fixed- K spread maps to the fixed- x one through the forward transformation $K \mapsto x = K/F_T$ and the discounting absorbed in the normalization (Note 06), so no separate condition is needed — but the two statements are related by that transformation, not identical coordinates.*

Proof sketch. Let $p = \partial_{xx} c$. Differentiating (1) twice in x gives the forward (Fokker–Planck-type) equation $\partial_T p = \frac{1}{2} \partial_{xx}(\nu x^2 p)$, whose initial datum is the Dirac mass of the payoff kink at $x = 1$ — nonnegative. Such equations preserve nonnegativity (parabolic maximum principle applied to the adjoint, or probabilistically: $p(T, \cdot)$ is the density of the diffusion $dX_t = \sqrt{\nu(t, X_t)} X_t dW_t$ started at 1). With $p \geq 0$ established, (ii) is immediate from (1): $\partial_T c = \frac{1}{2} \nu x^2 p \geq 0$. $\square$

The whole design of this model is to make proposition 1 survive discretization (section 4) and parametrization (section 3) — and then to *measure* what little the discretization leaks (section 4.3).

3 The piecewise-affine surface

3.1 Parametrization

Definition 1 (P_1 local-variance surface). On a tensor grid of $n_t \times n_x$ vertices (τ_i, ξ_j) the local variance is the continuous piecewise-affine interpolant

$$\nu_\theta(t, x) = \sum_{i,j} \theta_{ij} \phi_{ij}(t, x), \quad (2)$$

where the hat functions ϕ_{ij} are barycentric on a (cached Delaunay) triangulation of the vertices — each grid rectangle is split into two triangles, and the interpolant is genuinely affine on each — and $\theta_{ij} = \nu(\tau_i, \xi_j) > 0$ are the *nodal local variances*, the calibration parameters. The flat vector is time-major $\theta = \text{vec}(\theta)$.

(A bilinear tensor interpolant is kept for comparison, but bilinear is not affine on cells; the Delaunay barycentric basis is the golden convention.)

Proposition 2 (Nodal positivity is surface positivity). *On each triangle a P_1 function is a convex combination of its three vertex values. Hence $\theta_{ij} \geq \nu_{lo} > 0$ for all i, j implies $\nu_\theta(t, x) \geq \nu_{lo}$ everywhere on the grid’s hull, and the basis is a partition of unity (test-locked).*

This is why the calibration needs only *box bounds* on θ : positivity of a two-dimensional surface reduces to positivity of finitely many numbers. In production the box is $[\text{varLo}, \nu_{hi}]$ with $\text{varLo} = 0.0025$ (local vol 5%) and an *adaptive* cap $\nu_{hi} = \min(\max(0.36, (\text{lvVolCapMult} \cdot \sigma_{\max})^2), 16)$ — at least 60% vol, three times the largest observed implied vol if that is larger, hard-capped at 400%. (The tighter $[0.005, 0.20]$ box quoted in older revisions is only the model-layer function default used by the golden tests.)

Beyond the lowest strike vertex the surface extrapolates local variance linearly with slope `left_extrap_a` times the first interior cell’s slope. Three regimes: $a = 0$, the hot-path default, is a flat clamp; `convexWing=true` sets a to the *fixed* multiple `leftWingSlopeMult` (1.5) even without a variance-swap quote; and a becomes a *free* calibration variable, bounded $[0, 20]$, exactly when a var-swap quote is present (Note 08). The right wing is flat-clamped.

Caution: the extrapolation is outside the positivity guarantee

Proposition 2 stops at the triangulation hull. Outside it the basis carries signed weights and the linear continuation is unconstrained in both directions: with $a > 0$ and a first cell whose variance *decreases* toward $x_{\min}$, the extrapolated variance can go *negative* before $x = 0$ (pricing floors it, so the priced curve kinks rather than blows up); symmetrically, a rising continuation exceeds the nodal cap freely — the cap is deliberately not applied in the extrapolation region (`models/localvol/affine.py`). Every guarantee in this note that leans on $\nu \leq \nu_{hi}$ or $\nu > 0$ is therefore scoped to the hull (or to the flat-wing $a = 0$ path, where the clamp inherits the boundary vertex’s box); an extrapolation-domain positivity constraint is an open follow-up.

Remark 1 (Lee’s bound, on the hull). Bounded positive local variance bounds the total implied variance uniformly in strike — by comparison with the constant-variance diffusion, $w(k, T) \leq \nu_{hi} T$ — so the implied wing slope $dw/d|k|$ decays to zero, strictly inside Lee’s cap of 2 (Note 09). Scope this like everything above: the comparison holds where $\nu \leq \nu_{hi}$ holds, i.e. on the hull and on the flat-clamped wings; a steep free- a put continuation escapes the cap and with it this bound. Unlike the parametric slices no wing penalty is carried, and the affine surface currently exposes no per-fit Lee-slope fields — wing behaviour is a stated extrapolation contract plus the grid diagnostics of section 4.3, not a reported slope.

3.2 Grid design, and the short-dated stack

The strike vertices are delta-spaced (at the deltas $\{0.01, 0.02, 0.05, 0.10, 0.25, 0.40, 0.50\}$ and their call mirrors, sized by the *longest* expiry’s ATM std) and floored per expiry so a narrow short-dated smile still lands enough vertices on its curvature (`gridXNodes`, `gridXMinPerExpiry`; the case file of section 9 is exactly the failure that motivated the floor). The time vertices (`gridTNodes`) carry the term structure. The grid is deliberately modest — a dozen strike vertices, ten in time — because each vertex is a free parameter and PDE cost scales with the grid; the roughness regularizer (section 6) supplies the smoothness a coarse grid would otherwise lack.

The short end has since accreted a full rescue *stack* (the 2026-07-11 daily-ladder pass, which post-dates and extends the original case file):

1. **Adaptive variance box.** The cap is $\nu_{hi} = \min(\max(0.36, (\text{lvVolCapMult} \cdot \sigma_{\max})^2), 16)$ as above; the *floor* is now adaptive symmetrically, $\nu_{lo} = \min(\text{request floor}, (0.5 \min_i \sigma_{\text{ATM},i})^2)$ — a low-vol short-dated smile needs local vol *below* its minimum implied (implied vol is a path average), and the fixed 5% floor was measured riding the box on SPY 2-DTE upside quotes. Keyed on ATM implies deliberately, so a noisy deep-wing quote cannot drag the floor down where the box does real stabilization work.
2. **PDE strike refinement.** Δx refines to $0.15\times$ the shortest expiry's ATM $\sigma\sqrt{\tau}$ (was $0.3\times$), snapped so $x = 1$ stays a node, capped at 800 nodes (was 400): on 2-DTE dailies the *quote spacing* is finer than the lattice, and the drawn smile wiggled at quote frequency until the step out-resolved it.
3. **PDE time refinement.** Any short maturity interval that would receive fewer than 8 implicit steps at the $\Delta\tau = 0.01$ ceiling is marched with 32 steps instead (a 2-day interval otherwise gets *one*).
4. **Even-gap coverage.** For expiries up to ten days the per-expiry count floor is upgraded to an *even-gap* rule — keep splitting the widest boundary-augmented gap until none exceeds $\text{range}/(m_{\min} - 1)$ — because the count floor alone is side-blind: a downside-heavy axis met it with a single upside vertex and the daily front drew a V through its call quotes.
5. **Chained front tie.** When the first quoted expiry is below 0.08 years, the front tie chains over *every* vertex row before it (quotes pin only the variance integral to T_1 , so each sub-front row is individually unidentified) at an effective weight of at least 1 (up from 10^{-2}) — measured 5–30 vol-point ringing between sub-front rows collapses, and operator refinement becomes monotone.

Each rescue is gated on the regime that needs it; normal surfaces take none of these branches (locked byte-identical where a lock exists, e.g. the coverage floor and Δx gates).

4 The discrete pricing map

4.1 Discretization

The forward map (1) is solved on a uniform strike grid with step $\Delta x = 0.01$ on $x \in [0, x_{\max}]$, $x_{\max} \gtrsim 2.5$ (~ 250 nodes; the short-dated refinement of section 9 can tighten Δx to $1/400$), with Dirichlet boundaries $c(\cdot, 0) = 1$, $c(\cdot, x_{\max}) = 0$, by fully implicit (backward) Euler steps of size $\Delta T = 0.01$, the local variance entering at the *new* time level. On a (possibly non-uniform) grid with gaps $h_i^- = x_i - x_{i-1}$, $h_i^+ = x_{i+1} - x_i$, the operator A encoding $\frac{1}{2}\nu x^2 \partial_{xx}$ has the three-point stencil

$$A_{i,i\mp 1} = \nu_i a_i^\mp, \quad A_{ii} = -\nu_i(a_i^- + a_i^+), \quad a_i^\mp = \frac{x_i^2}{(h_i^- + h_i^+) h_i^\mp} > 0, \quad (3)$$

and each step solves the tridiagonal system

$$B^{n+1}U^{n+1} = U^n + (\text{boundary terms}), \quad B^{n+1} = I - \Delta T A^{n+1}. \quad (4)$$

Proposition 3 (The implicit step matrix is an M -matrix). *For any $\nu \geq 0$ and any $\Delta T > 0$, the matrix $B = I - \Delta T A$ of (4) satisfies: positive diagonal $B_{ii} = 1 + \Delta T \nu_i(a_i^- + a_i^+) \geq 1$, non-positive*

off-diagonals $B_{i,i\mp 1} = -\Delta T \nu_i a_i^\mp \leq 0$, and unit row sums at interior rows. Hence B is a strictly diagonally dominant nonsingular M -matrix, and $B^{-1} \geq 0$ elementwise.

Proof. The sign pattern and row sums are read off (3): $B_{ii} - |B_{i,i-1}| - |B_{i,i+1}| = 1 + \Delta T \nu_i (a_i^- + a_i^+) - \Delta T \nu_i (a_i^- + a_i^+) = 1 > 0$, so B is strictly diagonally dominant with positive diagonal and non-positive off-diagonals — the defining pattern of a nonsingular M -matrix, and M -matrices are inverse-positive: $B^{-1} \geq 0$ (see e.g. Berman–Plemmons [5], Theorem 6.2.3; concretely, Jacobi iteration for $Bu = v$ with $v \geq 0$ has nonnegative iterates and converges by dominance). $\square$

Corollary 1 (Discrete maximum principle and unconditional stability). *For any $\Delta T > 0$ the scheme (4) is monotone ($U^n \geq V^n$ with the same boundary data implies $U^{n+1} \geq V^{n+1}$) and satisfies $\min(\min_i U_i^n, c_{\text{bdry}}) \leq U_i^{n+1} \leq \max(\max_i U_i^n, c_{\text{bdry}})$: the implicit march is unconditionally ℓ^∞ -stable, with no CFL restriction, and cannot create new extrema.*

Remark 2 (What the M -matrix argument does and does not prove). Proposition 3 and corollary 1 give inverse positivity, order preservation and a maximum principle for the call values. They do not by themselves imply nonnegative second differences: a bounded, monotone call vector can still be locally concave, so “the scheme cannot manufacture negative densities” would need a separate convexity-preservation proof for the actual stencil and boundary elimination (a discrete Fokker–Planck argument), which this note does not carry. The honest statement is the one production implements: the scheme is monotone and stable, and butterfly and calendar cleanliness are *measured* on the output grid, per fit (section 4.3); the arbitrage-cleanliness tests in the suite are regression checks on selected surfaces, not a general proof.

Proof. Monotonicity is $B^{-1} \geq 0$ applied to $U^n - V^n \geq 0$. For the bounds, let M be the larger of $\max_i U_i^n$ and c_{bdry} : by the unit row sums $B(M\mathbf{1}) \geq M\mathbf{1}$ componentwise, and $M\mathbf{1}$ dominates the right-hand side of (4), so $U^{n+1} \leq M\mathbf{1}$ by inverse positivity; the lower bound is symmetric. $\square$

Strict diagonal dominance is also why no pivoting is ever needed: each step is a no-pivot Thomas march — the kernel the compiled march of appendix B vectorizes. (The scalar listing below is the *algorithm*; the production kernel `affine_march._march` is its multi-RHS vectorization with the sensitivity columns as the contiguous SIMD inner loop.)

Listing 1: The no-pivot tridiagonal (Thomas) solve for one implicit step (4) — the scalar algorithm that `models/localvol/affine_march.py` vectorizes; matches a dense solve to 10^{-10} .

```

1  def thomas(a, b, c, d):          # a: sub-, b: main, c: super-diagonal
2      n = len(d); cp = np.empty(n); dp = np.empty(n)
3      cp[0], dp[0] = c[0]/b[0], d[0]/b[0]
4      for i in range(1, n):       # forward sweep (no pivoting: Prop. M-
                                   matrix)
5          m = b[i] - a[i]*cp[i-1]
6          cp[i], dp[i] = c[i]/m, (d[i] - a[i]*dp[i-1])/m
7      x = np.empty(n); x[-1] = dp[-1]
8      for i in range(n-2, -1, -1):
9          x[i] = dp[i] - cp[i]*x[i+1]
10     return x

```

4.2 Why not Crank–Nicolson

Remark 3 (CN loses monotonicity exactly when it matters). Crank–Nicolson replaces (4) by $(I - \frac{\Delta T}{2}A)U^{n+1} = (I + \frac{\Delta T}{2}A)U^n$. The implicit factor is still an M -matrix, but the *explicit* factor has diagonal $1 - \frac{\Delta T}{2}\nu_i(a_i^- + a_i^+)$, which is nonnegative only under the CFL-like bound $\Delta T \leq 2/(\nu_i(a_i^- + a_i^+))$ — violated precisely on coarse strike grids with high local variance. When it is violated the scheme is second-order accurate but *not* monotone, and the payoff kink excites the classic CN oscillations. This is not hypothetical: the opt-in Rannacher/CN scheme (two implicit start-up steps damping the kink [4], then CN) bought only $\sim 1.1\times$ and produced a measured coarse- x arbitrage violation on a test surface — so the default stays fully implicit, trading one order of time accuracy for corollary 1.

4.3 Observation operator and no-arbitrage diagnostics

The observation operator reads the normalized call at each quote’s (τ, x) off the marched solution by interpolation. Cleanliness of the output is *measured*, with the production tolerances stated exactly: the per-expiry minimum second difference (butterfly proxy) must be $\geq -10^{-6}$, adjacent maturities may not decrease by more than 10^{-9} (calendar), and prices must sit in $[-10^{-9}, 1 + 10^{-9}]$ (bounds) — all three surfaced as diagnostics and an `arbitrageFree` flag. Two further layers: the fit quality is additionally repriced on a *converged* operator (`d $\tau$ /4, d $x$ /2; rmsConvergedBp`) because in-operator residuals are blind to time-discretization error the optimizer compensates; and a separate, finer log-strike Crank–Nicolson/Rannacher validator (`models/localvol/pde.py; atlas`) checks the production march against an independent discretization.

5 Variance swaps

A variance-swap quote is the fair strike of realized variance — for a diffusion, determined by the expected integrated local variance to expiry. Two distinct objects share the section: the *total* variance $I(\tau) = \mathbb{E} \int_0^\tau \nu(s, X_s) ds$, which is what the residual block prices, and the quoted *annualized fair strike* $I(\tau)/\tau$. Two pricers:

- **Static replication** (default): the log-contract weights

$$I(\tau) = 2 \int_0^1 \frac{P(\tau, x)}{x^2} dx + 2 \int_1^\infty \frac{C(\tau, x)}{x^2} dx, \quad (5)$$

integrated by trapezoid over the *PDE strike grid* itself — no separate replication grid — with the put leg starting at the *fixed* normalized strike $x = 0.01$ (`varswap_k_lo`; the x^{-2} weight diverges at zero), not at the lowest quoted strike.

- **Source PDE** (opt-in, `varSwapMethod=source_pde`): solve one *backward* PDE $\partial_t g + \frac{1}{2}\nu x^2 \partial_{xx} g + \nu = 0$, $g(\tau, \cdot) = 0$, and read $I(\tau) = g(0, 1)$ — the Feynman–Kac representation of $\mathbb{E} \int_0^\tau \nu(t, X_t) dt$. It reuses the same tridiagonal machinery marched backward and carries *analytic* sensitivities $\partial I / \partial \theta$ (and $\partial I / \partial a$ for the free left slope) through the same multi-RHS trick. Its selling point is stated carefully: $g(0, 1)$ is determined by the diffusion *around* $x = 1$, so it is materially less sensitive to the far strike boundary than the x^{-2} -weighted replication tail — but it still runs on the finite $[0, x_{\max}]$ domain with approximate boundary conditions, so “no truncation tail” would be too absolute. Note 08 develops both in full.

6 What the data identifies, and the objective

6.1 What the data identifies

Before writing the objective, be honest about the inverse problem. A typical fit has $m = n_t \times n_x \approx 100$ –200 nodal parameters (plus the free left slope a when a var-swap is quoted) against a few dozen to a couple of hundred vanilla quotes concentrated near the money of a handful of expiries — and even where counts balance, vanillas constrain local variance only through smoothing integrals along diffusion paths. Sparse vanillas therefore do *not* uniquely identify the grid: whole regions (deep wings, the sub-front rows, late times beyond the last expiry) are weakly determined, and what the calibration returns is one *regularized representative* of an equivalence class. Each regularizer picks a direction of that choice: the roughness block selects smoothness toward θ_{ref} , the box bounds clip the unidentified extremes, the (chained) front tie pins the sub-front rows to the first identified row, the prior baskets pull toward yesterday where quotes are silent, and the seed decides which basin a non-convex solve lands in. Low *quote* error consequently does not mean the latent surface was recovered — the synthetic example of section 10 now measures both numbers separately.

6.2 The objective

With unweighted residual blocks $r_\bullet$ (each defined below *without* its weight), the fit minimizes a stacked, bound-constrained least-squares residual over the decision vector — θ alone on the hot path, or (θ, a) with $a \in [0, 20]$ when the left slope is fitted (its column enters the Jacobian through the extrapolation basis):

$$\min_{\substack{\theta \in [\nu_{\text{lo}}, \nu_{\text{hi}}]^m \\ (a \in [0, 20])}} \frac{1}{2} \left(\|r_{\text{opt}}\|^2 + \|r_{\text{vs}}\|^2 + \|r_{\text{bsk}}\|^2 + \lambda \|L(\theta - \theta_{\text{ref}})\|^2 + w_{\text{cvx}} \|r_{\text{cvx}}\|^2 + w_{\text{ft}} \|r_{\text{ft}}\|^2 \right), \quad (6)$$

i.e. each weight appears *once*: production stacks the rows $\sqrt{w_\bullet} r_\bullet$, whose squared norm reproduces (6) exactly. The blocks:

- **Option residuals** $r_{\text{opt},q} = (P_q(\theta) - y_q)/\eta_q$ in vega-normalized forward-call units, with $\eta_q = 0.01 \max(\text{vega}, 10^{-3})/\sqrt{\omega_q}$; or the bid-ask/haircut band hinge with a small mid anchor of weight 0.05 (`midAnchorWeight`) when band edges are present (Note 07).
- **Var-swap residuals** $r_{\text{vs}} = (I(\theta) - z)/\zeta$ in total variance (Note 08). One scalar per quoted expiry: it constrains the *global tail integral* (5), biasing the deep-put mass in aggregate — it does not uniquely identify the deep-put tail’s shape.
- **Operator baskets** r_{bsk} : the prior-persistence operators (Note 13) enter as *signed baskets* of option prices — one residual per ATM/RR/BF operator, preserving the risk-reversal and butterfly coupling rather than scattering it into synthetic quotes.
- **Roughness** $L(\theta - \theta_{\text{ref}})$ at weight λ : L is the *spacing-aware* second difference (cell-width-normalized divided differences, reducing to $(1, -2, 1)$ on a uniform grid, exact curvature on a non-uniform one), with time-vs-strike balance `gridRegRho`. This is the smoothness that lets a coarse grid generalize.
- **Convex wing** (opt-in) at weight w_{cvx} : a one-sided hinge $\text{relu}(-D^2\sigma)$ on the nodal local *vols*, confined to vertices at or below the 5Δ put *and strictly below the deepest observed quote* — the

extrapolation tail only (section 9).

- **Front tie** (on by default) at weight w_{ft} : the one-sided differences $\theta_{i,:} - \theta_{i+1,:}$; tying the unidentified sub-front rows to the first data-identified expiry row — the single $t = 0$ row normally, the full chain at effective weight ≥ 1 on short fronts (section 3.2).

7 Sensitivities: tangent and adjoint

Everything the optimizer needs follows from differentiating the stepped system (4). Since ν_θ is linear in θ (2), so is $A(\theta)$, with $\partial A / \partial \theta_\ell = A[\phi_\ell]$ — the same stencil (3) with ν replaced by the basis function ϕ_ℓ .

Proposition 4 (Discrete tangent = same factor, extra right-hand sides). *The sensitivity $s_\ell^n = \partial U^n / \partial \theta_\ell$ obeys*

$$B^{n+1} s_\ell^{n+1} = s_\ell^n + \Delta T A[\phi_\ell] U^{n+1}, \quad s_\ell^0 = 0, \quad (7)$$

the discretization of the continuous sensitivity PDE $\partial_T s_\ell = \frac{1}{2} \nu x^2 \partial_{xx} s_\ell + \frac{1}{2} \phi_\ell x^2 \partial_{xx} c$. All m sensitivities share the same tridiagonal factor as the value solve, so the whole Jacobian is one factorization plus m extra back-substitutions per step — $O(N_T N_x m)$, and the reason the vectorized march carries the columns as its inner SIMD loop.

Proof. Differentiate $B^{n+1}(\theta) U^{n+1}(\theta) = U^n(\theta)$ in θ_ℓ : $B^{n+1} s_\ell^{n+1} + (\partial_{\theta_\ell} B^{n+1}) U^{n+1} = s_\ell^n$, and $\partial_{\theta_\ell} B^{n+1} = -\Delta T A[\phi_\ell]$ by linearity of A in θ . (Sensitivities against FD: test-locked at 10^{-6} .) $\square$

Proposition 5 (Discrete adjoint = one backward sweep). *Let $J(\theta) = f(HU^{1..N}(\theta))$ be any scalar objective reading the marched solution through observation rows H . Define adjoint states by the backward recursion*

$$(B^n)^\top \lambda^n = H_n^\top \partial f_n + \lambda^{n+1}, \quad \lambda^{N+1} = 0, \quad (8)$$

where $H_n^\top \partial f_n$ collects the objective's direct dependence on step n . Then

$$\frac{\partial J}{\partial \theta_\ell} = \sum_n \lambda^n^\top \Delta T A[\phi_\ell] U^n, \quad (9)$$

at cost one backward PDE solve, $O(N_T N_x)$, plus at least $O(m)$ output accumulation over (9) — the PDE cost independent of the parameter count m .

Proof. Form the Lagrangian $\mathcal{L} = f(HU) - \sum_n \lambda^n^\top (B^n U^n - U^{n-1})$ (boundary terms suppressed). Stationarity in each U^n gives (8); $\partial \mathcal{L} / \partial \theta_\ell$ then picks up only the explicit θ -dependence of B^n , which is $-\Delta T A[\phi_\ell]$, giving (9). This is summation-by-parts: the transpose of a lower block-bidiagonal system is upper block-bidiagonal, i.e. a backward march with the *transposed* tridiagonal factors. $\square$

What production actually runs is narrower than the theory above, and the distinction matters. Production marches *all* forward sensitivities (7), materializes the dense option Jacobian from them, and evaluates both products as matrix products: `apply_jacobian` is Jv , and the transpose product is literally $J^\top w$ on the assembled matrix (plus a sparse regularization block) — *not* the backward recursion (8). The matrix-free GN solver's win is therefore purely in the *linear algebra*: it avoids forming $J^\top J$ and the dense per-iteration SVD, but its data Jacobian is still assembled from forward sensitivities at $O(N_T N_x m)$ per evaluation. The true PDE adjoint of proposition 5 — which would

make gradient cost independent of m — is derived here as future theory and is *not currently implemented*. What is test-locked is the implemented pair: the product identity $\langle Jv, w \rangle = \langle v, J^T w \rangle$ and the gradient against finite differences.

8 Solvers and the speed story

The cold fit of (6) was, at ~ 533 vertices, an ~ 86 -second affair hitting the 200-evaluation cap — the heaviest single computation in the app. Driving it down is most of the engineering in this model, and the lessons are worth stating because several plausible ideas *failed*.

8.1 Where the time goes

On a cold fit the cost splits roughly as: optimizer linear algebra / dense SVD $\sim 52\%$, PDE sensitivity march $\sim 32\%$, Jacobian assembly $\sim 14\%$, value solve $\sim 2\%$. No single per-evaluation lever moves the total — which is why four separate “faster per evaluation” attempts (coarse grid, first Numba try, Rannacher, and the *first* matrix-free GN) each won only ~ 1.1 – $1.2\times$ and were shelved.

8.2 What worked

Performance

The cold fit is now ~ 3 – $6\times$ faster than the banded baseline, recalibrations near-instant. The levers, each with its honest scope:

1. **Stall-based early-stop.** Track the option-block misfit and stop at the best iterate once it stalls. Fewer evaluations scale march, assembly *and* optimizer together: measured $1.45\times$ (SPY, 200 $\rightarrow$ 109 evals) to $3.3\times$ (NVDA, 200 $\rightarrow$ 41) — at a *deliberately accepted* $+0.10$ – 0.25 vol bp drift from the full-cap fit. Window 0 is byte-identical.
2. **Compiled vectorized-Thomas march.** A @njit no-pivot Thomas with the sensitivity columns as the contiguous SIMD inner loop and the source fused into the forward sweep: 6.1 – $6.9\times$ the LAPACK banded march *march-only*, exact to $\sim 10^{-15}$. The first Numba attempt (column-outer scalar Thomas) won only $1.2\times$ — the lever was the *loop order*, not the compiler.
3. **Matrix-free Gauss–Newton** (default `lvSolver=gn`), scoped precisely: it engages only for the smooth *mid* fit with the Numba kernel active and no free- a var-swap path; band/haircut objectives, var-swap fits and the no-Numba path all run TRF. Where it engages: 1.3 – $1.65\times$ over TRF with an automatic TRF fallback on breakdown — and the public schema states the accepted trade openly: GN can converge to a slightly *different* real-data optimum, up to ~ 0.25 vol bp (often better). The locking tests compare GN and TRF surfaces with tolerances; they do not prove an identical optimum.
4. **Warm seeds.** Previous-surface seed on recalibration — $\sim 38\times$ on a *small synthetic ALPHA test*, qualitatively large on real chains — and a parametric-Dupire cold seed (1.3 – $1.8\times$). θ_{ref} is held fixed so the regularized *objective* is unchanged; a non-convex bound-constrained solve from a different start is still not guaranteed the same solution, and the lock is the objective identity plus the observed agreement, not a uniqueness theorem.

The order-of-magnitude wins are spent; the remaining levers are incremental.

Caution (historical, pre-Numba): where GN lost

This paragraph records the *first* GN verdict, before the compiled march existed, because its lesson generalizes even though its conclusion was superseded by the default-GN promotion above. At that time, removing the dense SVD made fits *slower* at ≤ 440 vertices, because the bottleneck was the march, not the SVD; the clean synthetic benchmark (zero residual, interior optimum) hid this — GN converged in 8 evaluations there but blew past the cap on stiff real data. Once the Numba march collapsed the per-evaluation cost, the SVD share dominated and the same GN won. The durable lessons: judge speed-ups on real chains, distrust a synthetic-only win — the same benchmark discipline later caught the convex-wing regression (section 9) that a synthetic test could never see.

9 Case files

Case file: the six-day weekly that broke the strike grid

Setup. A true 6-DTE SPY weekly (the Massive capture `lv_weekly_massive.json`; the Bloomberg fixture's shortest expiry is three weeks, which is why the bug survived the benchmark). Normal expiries fit at a few vol bps.

Failure. The weekly fit was catastrophic: 108 vol bp RMS against the parametric model's ~ 47 on the same quotes — the LV surface visibly missing the smile's curvature.

Diagnosis — measure first. A pure side-channel diagnostic (`affine_diag.py`) counts, per expiry: in-range strike vertices (Cause A), vega-floored quotes (Cause B), PDE time steps (Cause C), active prior rows (Cause E). The counts acquitted the usual suspects — time steps adequate, prior inactive, vega floor latent but not binding — and convicted Cause A: the delta strike axis is sized to the *longest* expiry and clipped to the global range, so the weekly's narrow smile landed *three* vertices on its sharpest curvature (fig. 1).

Fix. Two gated changes. (i) A per-expiry coverage floor (`gridXMinPerExpiry=8`): split the widest in-range gaps until every expiry has $\geq m_{\min}$ vertices inside *its own* traded range — densifying only under-covered short fronts. (ii) A short-expiry-aware PDE step: refine the shared Δx to a fraction of the smallest ATM $\sigma\sqrt{\tau}$, snapped so $x=1$ stays a node ($0.3\times$, capped at 400 nodes, at the time of this case; since tightened to $0.15\times/800$ by the daily-ladder pass of section 3.2).

Verdict. $108 \rightarrow \sim 28$ vol bp from the coverage floor alone (the short-end noise floor), 23.5 with the refined step — now *better* than the parametric fit — while every long expiry and every normal surface is byte-identical (both gates test-locked). This case is the historical seed of the short-end design, not its current entirety: the July-2026 daily-ladder stack of section 3.2 (adaptive floor, time refinement, even-gap coverage, chained tie) extends it.

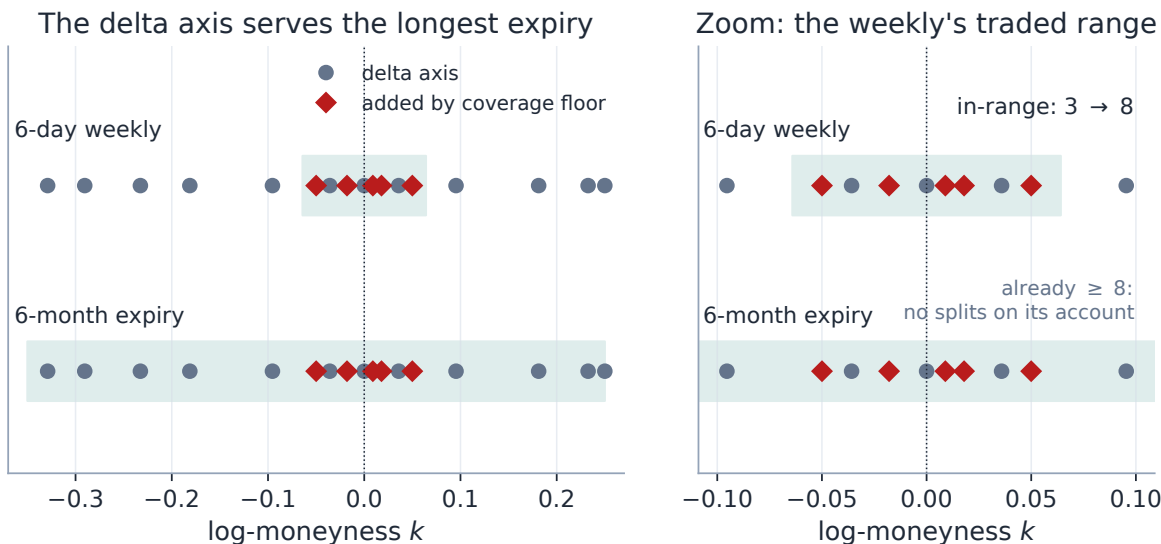


Figure 1: The short-dated rescue, computed by the production grid builder. The delta axis is sized by the longest expiry, so the six-day smile’s traded range (top lane) catches only 3 of its vertices; the coverage floor splits the weekly’s widest in-range gaps until it owns ≥ 8 . The six-month expiry already meets the floor, so no split is made on its account — the new vertices all land inside the weekly’s narrow range.

Case file: the convex wing that flattened SPY

Setup. The convex-wing hinge (section 6) exists to keep the *unquoted* deep-put extrapolation convex. A user ran with a saved `gridXNodes=20` — denser than the default 12.

Failure. SPY’s surface RMS degraded to 25.7 vol bp (the “26 bp” regression) on the Bloomberg benchmark; NVDA looked fine.

Diagnosis. The constraint originally selected *every* vertex at or below the 5Δ put, regardless of data. At 20 strike nodes several of those vertices sit *inside* SPY’s densely quoted put wing, and the hinge fought the quotes — forcing the wrong wing on a low-vol name. NVDA’s naturally convex wing had hidden the bug, and so had the default grid: only the denser grid pushed constrained vertices into quoted territory.

Fix. Confine `convex_cols` to vertices at or below the 5Δ put *and strictly below the deepest observed quote* — the extrapolation tail only.

Verdict. SPY 25.7 $\rightarrow$ 2.6 vol bp, NVDA unchanged; the confinement is test-locked. The lesson echoes Notes 05 and 09: *a shape constraint must not be imposed where the data already speaks* — the same principle, at the de-Am input, the MCS output, and here the LV extrapolation tail.

10 Worked examples

All numbers here carry provenance: regenerated by `figures/gen_lv.py` at commit 77d43fb (2026-07-11), with configuration, runtimes and library versions stored in `figures/lv_numbers.json`.

10.1 Synthetic quote recovery — and what it does not show

A known skewed local-variance surface is priced through the forward PDE, quotes are generated at four expiries, and a fit is run from a *flat* seed — through the *low-level* calibrator (TRF, banded march, custom dense grids): an algorithm check, not the product path or its timing. Figure 2 shows the result, and per section 6.1 we report two different numbers. The *quotes* reprice to a maximum of 0.6 vol bp in 33 evaluations (0.94 s wall, 28 vertices), and the broad skew is reconstructed. Separately, the *local-vol surface itself* agrees with the truth to 0.06 vol points RMS (0.25 max) on a dense grid over the quote-covered region — gratifyingly small here because 44 clean quotes constrain 28 vertices of a smooth truth, but it is the measured surface error, not an inference from the quote error, and it degrades off the covered region and on sparse real chains.

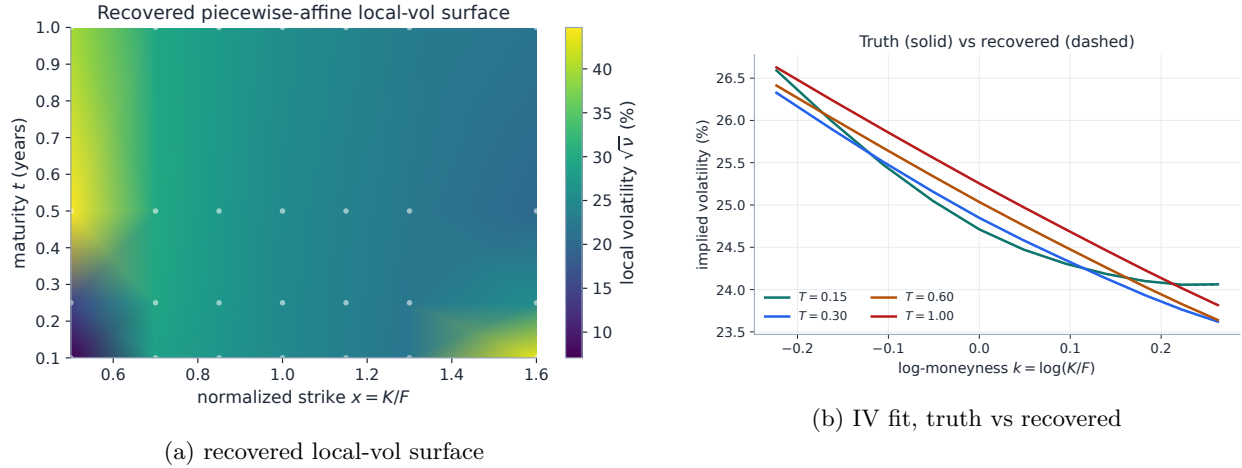


Figure 2: Synthetic recovery through the low-level calibrator: quotes reprice to 0.6 vol bp; the surface itself agrees to 0.25 vol points (max) on the quote-covered region — two separately measured numbers.

10.2 The Bloomberg benchmark

On the static Bloomberg fixture the *product-path* fit (`calibrate_affine_surface`, shipped request defaults) reads, in *two* metrics per name (table 1, fig. 3): the in-operator surface RMS — 2.4 vol bp on SPY, 11.9 on NVDA — and the *converged-operator* reprice ($d\tau/4$, $dx/2$): 15.9 and 56.3. The gap is the point: in-operator residuals are blind to time-discretization error the optimizer compensates, so the converged number is the honest fit quality, and closing that gap (per-expiry $d\tau$ refinement on fronts like NVDA’s) is the open short-end follow-up. The harder NVDA short expiries carry the larger residual in both metrics — the regime the short-dated stack of section 3.2 addresses.

Table 1: Product-path local-vol fit on the Bloomberg fixture (fresh run, commit 77d43fb). “Worst expiry” is in-operator.

Name	in-op RMS (bp)	converged RMS (bp)	worst expiry (bp)
SPY	2.4	15.9	3.4
NVDA	11.9	56.3	12.4

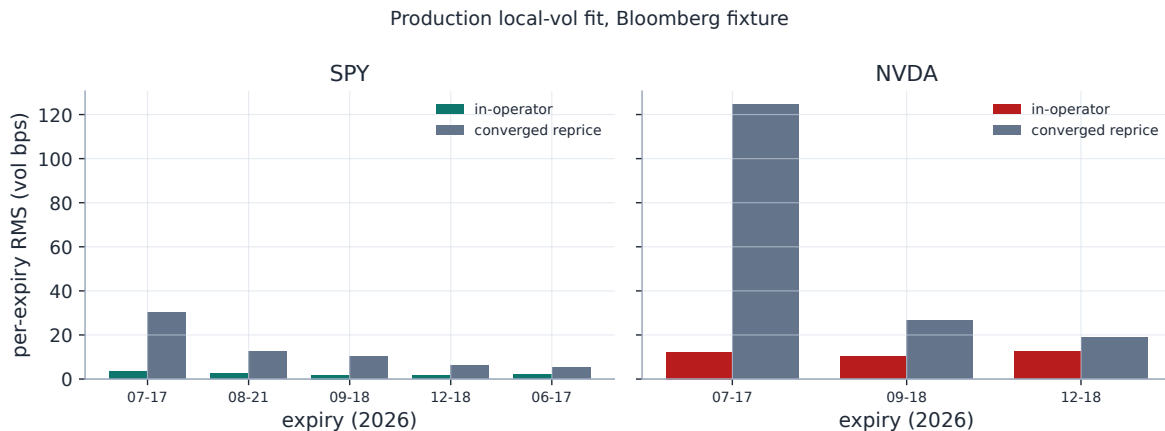


Figure 3: Per-expiry RMS for SPY and NVDA on the Bloomberg fixture — in-operator bars beside the converged-operator reprice, per name (expiry year in each panel’s axis label). The short end is where the residual lives in both metrics, and the converged bars are the honest ones.

11 What is original, and limitations

The P_1 -positivity guarantee and forward-calibration philosophy are Andreasen–Huge; the contributions here are depth and discipline. The *vectorized-Thomas Numba march* (loop order as the lever, $6.5\times$ march-only); the *stall early-stop* that scales the whole fit; the *matrix-free GN* first rejected, then correctly promoted when the march made the SVD dominant — consuming the forward-assembled Jacobian through the two matrix products of section 7; the *source-PDE var-swap* with analytic sensitivities; and the *measure-first* short-dated diagnosis that convicted the strike grid on counts rather than hunches. Each was validated against a real benchmark that twice caught regressions a synthetic test would have missed.

Two clarifications of scope. First, this note’s object is the *direct affine fit* — the surface calibrated to quotes. The application also carries a separate `/localvol` path that *extracts* a local-variance grid from the displayed parametric slices via the Dupire formula of appendix C; it is not merely a validator — the exact sticky-local-vol scenario transport currently prices off that extracted grid — but it is a different object with different error characteristics. Second, limitations: a pure diffusion cannot make a true short-dated event spike (Note 11’s variance clock absorbs scheduled events instead); the tensor grid wastes vertices in the corners (the future non-tensor bowtie grid is where the GN no-SVD advantage will finally dominate); the true PDE adjoint remains unimplemented (section 7); the extrapolation region sits outside the positivity guarantee (section 3); the converged-reprice gap on short fronts is open (section 10); and the fit, though much faster, remains the heaviest computation in the app.

12 Traceability

Read the anchors for what they lock. The “arbitrage-clean” tests are *regression checks on selected surfaces*, not proofs of remark 2’s missing convexity theorem; the “adjoint” tests lock the *implemented* matrix products of section 7, not the unbuilt backward-PDE recursion; the GN–TRF test compares surfaces *within tolerance* (the schema’s stated ~ 0.25 vol bp trade), not to identity; the warm-start

test locks the unchanged *objective* plus observed agreement; and the $108 \rightarrow 23.5 / 25.7 \rightarrow 2.6$ case-file improvements are prose evidence from the incident logs — the cited tests lock the *gates* (confinement, short-front-only densification), not those exact numbers.

Table 2: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor <i>Test anchors</i>
P_1 basis: partition of unity; nodal bounds bound the surface (hull)	proposition 2	models/localvol/affine.py test_localvol_affine.py::test_basis_partition_of_unity_and_nodal_interpolation ::test_nodal_bounds_imply_surface_bounds
Selected calibrated surfaces measure arbitrage-clean (regression, not proof)	remark 2	models/localvol/affine.py test_localvol_affine.py::test_calibrated_surface_is_arbitrage_free test_api_affine.py::test_affine_density_is_clean_no_interior_zeros
Scheme converges (flat + skew round trips)	section 4	models/localvol/pde.py test_localvol.py::test_spatial_convergence ::test_skew_dupire_round_trip
Tangent sensitivities exact vs FD	proposition 4	models/localvol/affine.py test_localvol_affine.py::test_forward_sensitivities_match_finite_differences
Implemented $Jv/J^T w$ products and gradient correct (matrix form, not the PDE adjoint)	section 7	models/localvol/affine_gn.py test_affine_gn.py::test_jacobian_transpose_inner_product_identity ::test_gradient_alpha_test
GN matches TRF within tolerance; fallback-then-stall safe (R1)	section 8	models/localvol/affine_gn.py test_affine_gn.py::test_gn_matches_trf_on_golden ::test_gn_trf_fallback_then_stall_returns_surface
Numba march = banded march	listing 1	models/localvol/affine_march.py test_affine_march.py::test_numba_march_matches_banded_prices_and_sens ::test_numba_calibration_matches_banded_surface
Convex wing confined to the extrapolation tail	section 9	api/affine_fit.py test_affine_grid_design.py::test_convex_wing_confined_to_quoted_extrapolation
Coverage floor / PDE step touch only short fronts	section 9	api/affine_fit.py test_affine_grid_design.py::test_coverage_floor_densifies_only_the_short_front ::test_pde_dx_refines_only_short_surfaces

Claim	Object	Code anchor <i>Test anchors</i>
Warm start: objective unchanged, evals cut	section 8	api/affine_fit.py test_affine_warm_start.py::test_recalibration_warm_starts_and_cuts_evals
Source-PDE var swap + free left slope	section 5	models/localvol/varswap_pde.py test_varswap_source.py test_affine_grid_design.py::test_free_a_reduces_varswap_error
Diagnostics are a pure side-channel	section 9	api/affine_diag.py test_affine_diag.py::test_record_shape_and_purity
Dupire extraction returns NaN on arbitrage	appendix C	models/localvol/dupire.py test_localvol.py::test_dupire_arbitrage_returns_nan
Prior operators enter as signed baskets	section 6	api/affine_fit.py, affine_calib.py test_affine_basket.py::test_basket_pulls_surface_toward_target ::test_empty_baskets_byte_identical

A Hyperparameter atlas

Table 3: Local-volatility hyperparameters.

Knob	Default	Role
<i>Surfaced (OptionsSettings)</i>		
gridStrikeMode	delta	Strike-vertex placement (delta-spaced).
gridXNodes	12	Strike vertices of the variance surface.
gridXMinPerExpiry	8	Min in-range strike vertices per expiry (section 9).
gridTNodes	10	Maturity vertices.
gridRegLambda λ	10^{-2}	Roughness strength (model-layer signature default is 10^{-4} ; production passes this).
gridRegRho ρ	1.0	Time-vs-strike roughness balance.
convexWing	/ false / 10^3	Convex-wing hinge (extrapolation tail only).
convexWingWeight		
frontTie	/ true / 10^{-2}	Front-end tie of the $t = 0$ row to the first expiry row.
frontTieWeight		
lvVolCapMult	3.0	Adaptive variance cap $\min(\max(0.36, (\text{mult} \cdot \sigma_{\max})^2), 16)$ — at most 400% vol.
leftWingSlopeMult	1.5	Left local-variance extrapolation slope multiple.
varSwapMethod	static	Var-swap pricer: replication over the PDE grid, or source-PDE.

Knob	Default	Role
lvSolver	gn	Matrix-free Gauss–Newton (default) or <code>trf</code> .
lvFastKernel	true	Numba vectorized-Thomas march.
lvEarlyStop	true	Stall-based early stop.
timeScheme	implicit	Implicit Euler (monotone) or opt-in Rannacher/CN (remark 3).
midAnchorWeight	0.05	Mid anchor inside the band objective.
<i>Hidden (affine PDE / grid / solver)</i>		
varLo / varHi	0.0025 / 0.36	Request-level nodal variance box (vol 5%/60%) before the adaptive cap <i>and floor</i> .
_LV_VOL_FLOOR_FRAC	0.5	Adaptive floor: $\nu_{lo} = \min(\text{request}, (0.5 \min \sigma_{ATM})^2)$ (section 3.2).
_X_DX / _X_MAX_MIN	0.01 / 2.5	PDE strike step and minimum $x_{\max}$.
_DT_MAX	0.01	Implicit-Euler maturity step (0.03 under Rannacher).
rannacher_steps	2	Implicit start-up steps before CN (opt-in scheme).
_PDE_DX_SHORT_FRAC /	0.15 / 800	Short-dated step: $0.15 \sigma \sqrt{\tau}$, capped at 800 nodes (was 0.3/400 before the daily-ladder pass).
_PDE_N_MAX		
_PDE_NT_FIRST_GATE /	8 / 32	Short-interval time refinement: 32 steps whenever the flat ceiling would give < 8 .
_PDE_NT_SHORT		
_COVERAGE_GAP_MAX_T	10/365	Expiry age below which the even-gap coverage rule applies (section 3.2).
FRONT_TIE_SHORT_T /	0.08 / 1.0	Chained front tie: threshold and minimum effective weight.
FRONT_TIE_CHAIN_WEIGHT		
varswap_k_lo	0.01	Fixed lower strike of the static replication’s put leg.
_DELTA_SET	7 deltas	Put deltas $\{1, 2, 5, 10, 25, 40, 50\}\%$ (and call mirrors) spacing the strike axis.
_CONVEX_WING_DELTA	0.05	Wing boundary for the convex hinge (5Δ put).
_VOL_TOL / _VEGA_FLOOR	$0.01 / 10^{-3}$	Vega normalization of option residuals.
diag tolerances	$-10^{-6} / 10^{-9}$	Butterfly proxy floor; calendar/bounds tolerances (section 4.3).
x_scale / tolerances	jac / 10^{-8}	Optimizer scaling; <code>max_nfev</code> = 200.
_STALL_WINDOW/_RTOL	12 / 5×10^{-3}	TRF early-stop ($18 / 3 \times 10^{-3}$ under GN).
VAR_FLOOR	10^{-6}	Dupire-extraction local-variance floor.
<i>Hidden (separate log-strike validator, <code>pde.py</code>)</i>		
DEFAULT_N_K /	1201 / 1/400	Validator grid.
DEFAULT_DT_MAX		
N_RANNACHER	4	Validator Rannacher quarter-steps.
SPAN_SD / SPAN_MIN	7.5 / 0.6	Validator grid span (sd / floor).

B Performance notes

1. **Numba march** (`lvFastKernel`). Factor-once no-pivot Thomas (proposition 3 is why no pivoting), m sensitivity columns as the contiguous inner SIMD loop, fused source: $6.1\text{--}6.9\times$ vs LAPACK banded, exact to $\sim 10^{-15}$; graceful banded fallback without Numba.
2. **Early stop** (`lvEarlyStop`). Best-iterate stop on option-block stall: $1.45\times$ (SPY) to $3.3\times$ (NVDA) at $+0.10\text{--}0.25$ vol bp; window 0 is byte-identical.
3. **Matrix-free GN** (`lvSolver=gn`). Preconditioned LSMR on the tangent/adjoint products of section 7; no dense SVD. $1.3\text{--}1.65\times$ over TRF at tensor sizes; golden-locked to the TRF optimum; automatic TRF fallback; sparse CSR regularization block.
4. **Warm starts**. Previous-surface recalibration $\sim 38\times$; parametric-Dupire cold seed $1.3\text{--}1.8\times$; θ_{ref} fixed so the optimum is unchanged.
5. **Solver scaling and tolerances**. `x_scale='jac'` plus termination tolerances relaxed $10^{-12} \rightarrow 10^{-8}$ on the TRF path: the fit is governed by quote noise, vega floors and bands, so the tighter tolerance only bought iterations — evaluation count cut, surface identical (the atlas records the resulting defaults).
6. **Shelved (documented in the perf roadmap)**: coarse-grid calibration (biases θ by up to ~ 26 vol points — non-viable), first Numba attempt (wrong loop order, $1.2\times$), Rannacher/CN ($\sim 1.1\times$ and a measured coarse- x arbitrage — remark 3), GN for the non-smooth band objective, thread/process parallelism (GIL-negative). Five dead-ends confirming the cost is distributed, not localized.

Caution (engineering): the field crash that pretended to be a matrix

The one production crash in this machinery was an interface bug, not a math bug: `LinearizedJacobian` deliberately has no `.T` attribute (it is an operator, not a matrix), and the early-stop handler, reached only on the rare GN $\rightarrow$ TRF fallback path that then stalls, called `jac.T @ r` to compute the final gradient — an `AttributeError` in the field (the R1 backtest finding). The fix dispatches to `apply_jacobian_transpose`, and the exact fallback-then-stall path is now test-locked. Lesson: an object that *pretends* to be a matrix must either implement the whole contract or make each unsupported member fail at construction, not mid-fit.

C Reference implementation: Dupire extraction

Forward calibration never differentiates the data (section 2), but the *inverse* map — reading a local-variance grid off an already-fitted total-variance surface $w(k, T)$ — is Gatheral’s formula [3], used for grid export, the parametric cold seed of section 8, and the graph LV projection (Note 14). The denominator g is the butterfly function: $g \leq 0$ marks strike arbitrage in the implied surface, where no positive local variance can reproduce it — returned as `nan`, never clipped. Matches production to 10^{-13} .

Listing 2: Dupire/Gatheral local variance from an implied total-variance surface (distilled from `models/localvol/dupire.py`).

```
1 def dupire_local_variance(k, w, wk, wkk, wt):
```

```

2      r = k/w                                     # w = w(k,T), with k/kk/T
      derivs
3      g = 1 - r*wk + 0.25*(-0.25 - 1/w + r*r)*wk*wk + 0.5*wk*wk
4      return np.where((w > 0) & (g > 0),          # g <= 0: butterfly arbitrage
      -> nan
5                                     wt/np.where(g > 0, g, 1.0), np.nan)

```

References

- [1] B. Dupire. Pricing with a smile. *Risk*, 7(1):18–20, 1994.
- [2] J. Andreasen and B. Huge. Volatility interpolation. *Risk*, March 2011.
- [3] J. Gatheral. *The Volatility Surface*. Wiley, 2006.
- [4] M. Giles and R. Carter. Convergence analysis of Crank–Nicolson and Rannacher time-marching. *J. Comput. Finance*, 9(4), 2006.
- [5] A. Berman and R. Plemmons. *Nonnegative Matrices in the Mathematical Sciences*. SIAM, 1994.